Chapter 7

Discrete random variables and binomial distri-ale

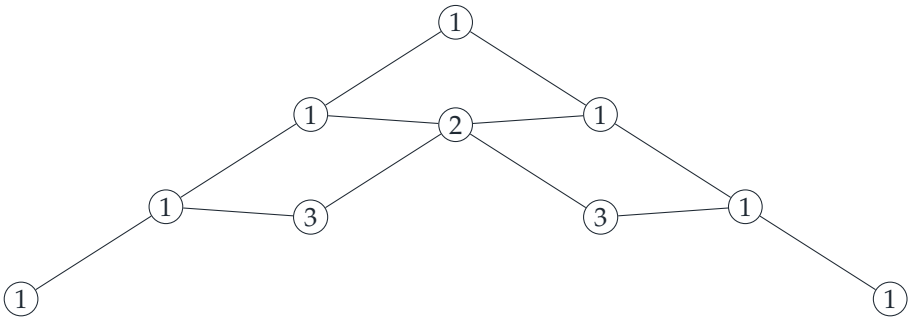
Skills: define a distribution, calculate and interpret an expectation, recognise and use a binomial distribution

7.1 Reference notes

A discrete random variable X assigns a real number to each outcome. Its distribution gives the values x_i and the probabilities $p_i = P(X = x_i)$, with $p_i \geq 0$ and $\sum p_i = 1$.

Key ideas

The expectation is $E(X) = \sum_i x_i p_i$. It represents the long-run mean value when the experiment is repeated under the same conditions; it is not necessarily a value actually taken by X.

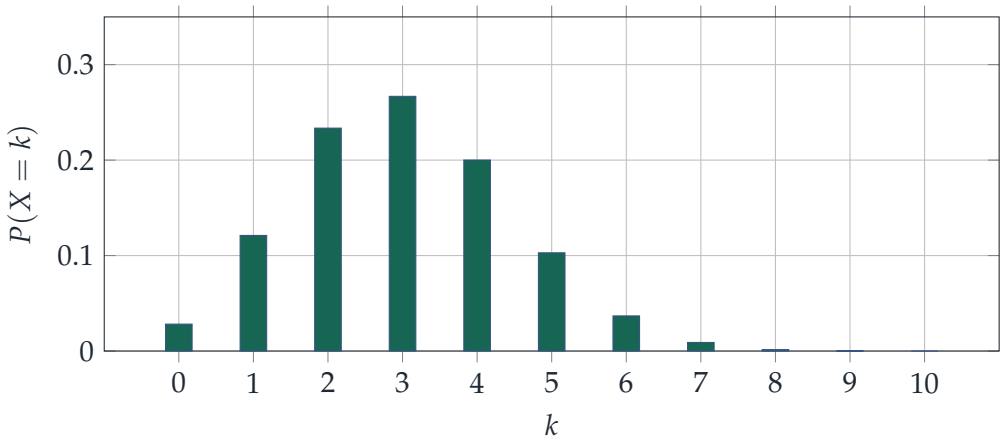


Key ideas

X follows a binomial distribution $\mathcal{B}(n, p)$ when X counts the number of successes in n trials of independent Bernoulli trials, with the same success probability p. Then

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad E(X) = np.$$

The coefficient $\binom{n}{k}$ counts the number of paths containing exactly k successes among n trials. For $n \leq 10$, it can be read in Pascal's triangle.



The diagram represents approximately $\mathcal{B}(10, 0{,}3)$; its centre is close to $E(X) = 3$.

Session 47 — Expectation of a random variable

Discrete random variables

06/04/2027 • B : 2 h – C : 1 h

Objectives of the session

· Define a distribution.

· Calculate an expectation.

· Interpret a theoretical mean.

Fluency · 6 to 8 minutes, without calculator

1. Simplify $x + 8 - x + 2$. $2x + 10$
2. Give $E(X)$ for $X \sim \mathcal{B}(10, 0{,}3)$. 3
3. Calculate $\binom{5}{2}$. 10
4. Calculate the expectation of X constant equal to 8. 8
5. Express $X \geq 1$ by a complement. $1 - P(X = 0)$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: define a distribution and calculate an expectation.

Investigation

A game costs 2 euros. The gross winnings are 0, 3 or 10 euros with probabilities 0 ,6, 0,3, 0,1. Construct the distribution of the net gain X, calculate $E(X)$ and decide whether the game is favourable to the player.

Detailed solutions · teacher edition

X is $-2, 1, 8$. $E(X) = -1{,}2 + 0{,}3 + 0{,}8 = -0{,}1$ euro: unfavourable game on average. Have students verbalise the piece of information that triggers the choice of method.

Written summary

The distribution of X assigns to each value x_i probability p_i. The expectation $E(X) = \sum x_i p_i$ is a mean theoretical value in the long run.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

X is $-2, 1, 8$. $E(X) = -1{,}2 + 0{,}3 + 0{,}8 = -0.1$ euro: unfavourable game on average.

Practice · essentials

1. Justify that the number of successes among 8 independent trials, with $p = 0{,}3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Detailed solutions · teacher edition

$X \sim \mathcal{B}(8, 0{,}3)$. $P(X = 2) = \binom{8}{2} 0{,}3^2 0{,}7^6$, $P(X = 0) = 0{,}7^8$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{24}{10}$.

References in the textbook

Extensions: Calao, notes pp. 154-159; exercises p. 160, nos. 1 to 5. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 160, Exercise 2

A game consists of rolling a fair die. If the result is:

· 1, 2 or 3, then you lose 2 points;

· 3 or 4, then you neither lose nor gain anything;

· 6, then you gain 3 points.

Let G the random variable which, to each roll, assigns the player's gain (positive or negative).

1. Complete the table that represents the probability distribution of the random variable G.
2. Calculate $E(G)$. What can we notice?

Detailed solutions · teacher edition

The printed text shows result 3 on two lines; the consistent partition is 1, 2, 3 for -2, 4, 5 for 0, and 6 for 3. We therefore obtain the probabilities $1/2$, $1/3$ and $1/6$, then $E(G) = -2 \times 1/2 + 3 \times 1/6 = -0.5$. The game is unfavourable to the player on average.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 355 and the rate or common ratio by 7. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of exercises essentials; 5 min of correction; 3 min of exit ticket.
Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.
Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 48 — Binomial distribution: recognising a situation

Discrete random variables

08/04/2027 • B : 1 h – C : 2 h

Objectives of the session

· Recognise a Bernoulli scheme.

· Identify n and p.

· Define X.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 180. 9

2. Calculate $\binom{5}{2}$. 10

3. Calculate the expectation of X constant equal to 9. 9

4. Express $X \geq 1$ by a complement. $1 - P(X = 0)$

5. Give $E(X)$ for $X \sim \mathcal{B}(10, 0{,}3)$. 3

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: recognise a Bernoulli scheme and identify n and p.

Investigation

A customer accepts an offer with probability 0 ,25. An adviser independently contacts eight customers. Define X, justify the binomial distribution and identify n and p.

Detailed solutions · teacher edition

X counts acceptances; repetition of 8 identical and independent trials, success of probability 0.25: $X \sim \mathcal{B}(8, 0{,}25)$. Have students verbalise the information that triggers the choice of method.

Written summary

$X \sim \mathcal{B}(n, p)$ if X counts the successes in n independent trials with the same probability p.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

X counts acceptances; repetition of 8 identical and independent trials, success of probability 0.25: $X \sim \mathcal{B}(8, 0{,}25)$.

Practice · essentials

1. Justify that the number of successes among 9 independent trials, with $p = 0{,}3$, follows a distribution binomial.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Detailed solutions · teacher edition

$X \sim \mathcal{B}(9, 0{,}3)$. $P(X = 2) = \binom{9}{2} 0{,}3^2 0{,}7^7$, $P(X = 0) = 0{,}7^9$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{27}{10}$.

References in the textbook

Extensions: Calao, notes pp. 154-159; exercises p. 160, no. 5, then p. 162, nos. 14 to 20. The exercise reproduced in this session is identified below by its page and number; the reference allows also to find the figures and digital resources of the textbook.

Calao textbook · p. 163, Exercise 22

The mayor of a town has decided to hold a referendum on the possible development of a park. The results are as follows: 63% for and 37% against. Two ballot papers are drawn at random of the vote. The number of participants is large enough for this to be treated as sample with a random draw, with replacement, of two ballot papers.
Consider the random variable X which, to any sample of two ballots, assigns the number of "For" ballots.

1. What is the distribution of X ? Determine its parameters.
2. Calculate the probability that such a sample contains: no "For"; exactly one " For".
3. Deduce from question 2 the probability that such a sample contains at most one "For ».

Detailed solutions · teacher edition

$X \sim \mathcal{B}(2; 0{,}63)$. We have $P(X = 0) = 0{,}37^2 = 0{,}1369$ and $P(X = 1) = 2 \times 0{,}63 \times 0{,}37 = 0{,}4662$. Thus $P(X \leqslant 1) = 0{,}6031$.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 360 and the rate or common ratio by 2. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of exercises essentials; 5 min of correction; 3 min of exit ticket.
Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.
Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 49 — Pascal's triangle and binomial coefficients

Discrete random variables

27/04/2027 • B : 2 h – C : 1 h

Objectives of the session

· Build Pascal's triangle.

· Read $\binom{n}{k}$.

· Interpret a coefficient.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier for a 10% increase. $1+\dfrac{10}{100}=1,10$
2. Calculate the expectation of X constant equal to 3. 3
3. Express $X \geq 1$ by a complement. $1-P(X=0)$
4. Give $E(X)$ for $X \sim \mathcal{B}(10,0{,}3)$. 3
5. Calculate $\binom{5}{2}$. 10

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: build Pascal's triangle and read $\binom{n}{k}$.

Investigation

Construct the rows $n=0$ à $n=6$ of Pascal's triangle. Explain how the number 15 in the line 6, then read $\binom{6}{2}$ and $\binom{6}{4}$.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

Detailed solutions · teacher edition

Each inner number is the sum of the two numbers above. $\binom{6}{2}=\binom{6}{4}=15$. Have students put into words the information that triggers the choice of method.

Written summary

Pascal's triangle satisfies $\binom{n}{k}=\binom{n-1}{k-1}+\binom{n-1}{k}$. The edges equal 1.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

Each inner number is the sum of the two numbers above. $\binom{6}{2} = \binom{6}{4} = 15$.

Practice · essentials

1. Justify that the number of successes among 10 independent trials, with $p = 0{,}3$, follows a distribution binomial.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Detailed solutions · teacher edition

$X \sim \mathcal{B}(10, 0{,}3)$. $P(X = 2) = \binom{10}{2} 0{,}3^2 0{,}7^8$, $P(X = 0) = 0{,}7^{10}$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{30}{10}$.

References in the textbook

Extensions: Calao, p. 164, nos. 24 to 29, then p. 166, no. 36. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 161, Exercise 13

A fair coin is tossed 5 times in a row. Let F the random variable which, to each experiment thus defined, assigns the number of "Heads" obtained.

1. F follows a binomial distribution. State its parameters.
2. Build Pascal's triangle up to $n = 5$.
3. Calculate $P(F = 3)$. Interpret this result.
4. Calculate the probability that "Heads" appears at least three times.

Detailed solutions · teacher edition

$F \sim \mathcal{B}(5; 1/2)$. The line $n = 5$ of the triangle is 1, 5, 10, 10, 5, 1. Hence $P(F = 3) = \binom{5}{3}(1/2)^5 = 10/32 = 0.3125$ and $P(F \geqslant 3) = (10 + 5 + 1)/32 = 0{,}5$.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 365 and the rate or common ratio by 3. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of exercises essentials; 5 min of correction; 3 min of exit ticket.
Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.
Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 50 — Binomial probabilities

Discrete random variables

29/04/2027 • B : 1 h – C : 2 h

Objectives of the session

· Calculate $P(X = k)$.

· Translate at least/at most.

· Use the complement.

Fluency · 6 to 8 minutes, without calculator

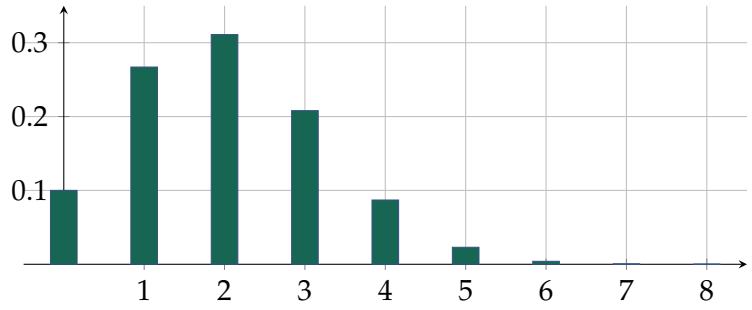
1. Simplify 3 $x + 4 - x + 2$. $2x + 6$
2. Express $X \geq 1$ by a complement. $1 - P(X = 0)$
3. Give $E(X)$ for $X \sim \mathcal{B}(10, 0{,}3)$. 3
4. Calculate $\binom{5}{2}$. 10
5. Calculate the expectation of X constant equal to 4. 4

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: calculate $p(x = k)$ and translate at least/at most.

Investigation

For $X \sim \mathcal{B}(8, 0{,}25)$, calculate $P(X = 2)$, $P(X = 0)$ then $P(X \geq 1)$. Express each event as sentence.



Detailed solutions · teacher edition

$P(X = 2) = \binom{8}{2} 0{,}25^2 0{,}75^6 \approx 0{,}3115$; $P(X = 0) = 0{,}75^8 \approx 0{,}1001$; $P(X \geq 1) \approx 0{,}8999$. Have students put into words the information that triggers the choice of method.

Written summary

If $X \sim \mathcal{B}(n, p)$, then $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$. Carefully express "at least", "at most " and "none".

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

$P(X=2)=\binom{8}{2}0{,}25^2 0{,}75^6 \approx 0{,}3115$; $P(X=0)=0{,}75^8 \approx 0{,}1001$; $P(X\geq 1)\approx 0{,}8999$.

Practice · essentials

1. Justify that the number of successes among 6 independent trials, with $p = 0{,}3$, follows a distribution binomial.
2. Calculate $P(X=2)$ and $P(X=0)$.
3. Deduce $P(X\geq 1)$ and calculate $E(X)$.

Detailed solutions · teacher edition

$X \sim \mathcal{B}(6,0{,}3)$. $P(X=2)=\binom{6}{2}0{,}3^2 0{,}7^4$, $P(X=0)=0{,}7^6$, $P(X\geq 1)=1-P(X=0)$ and $E(X)=\dfrac{18}{10}$.

References in the textbook

Extensions: Calao, p. 164, nos. 24 to 29, then p. 168, nos. 41 to 45. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 167, Exercise 38

A parcel delivery company manages, for parcels collected before 10:00, to deliver them as early as the next day with probability 0 ,75. Before 10 a.m., a shopkeeper drops off seven parcels for his customers. Let X the random variable which, to each batch of seven parcels, assigns the number of parcels delivered the next day. We accept that X follows a binomial distribution.

1. Determine the parameters n and p of the distribution X.
2. Calculate the probability of the following events: A: "All the parcels will arrive the next day." ; B : "At least one parcel will not arrive the next day."
3. Complete Pascal's triangle up to row $n = 7$. Deduce $\binom{7}{2}$. Calculate the probability of event C: "Exactly two parcels will arrive the next day."
4. Interpret the event $\{X=4\}$. Calculate the probability $P(X=4)$.

Detailed solutions · teacher edition

$X \sim \mathcal{B}(7;0{,}75)$. $P(A)=0{,}75^7 \approx 0{,}1335$ and $P(B)=1-0{,}75^7 \approx 0{,}8665$. Since $\binom{7}{2}=21$, $P(X=2)=21(0{,}75)^2(0{,}25)^5 \approx 0{,}0115$. $\{X=4\}$ means that exactly four packages are delivered on next day and $P(X=4)=\binom{7}{4}(0{,}75)^4(0{,}25)^3 \approx 0{,}1730$.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 370 and the rate or common ratio by 4. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of exercises essentials; 5 min of correction; 3 min of exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 51 — Chapter review and assessment

Discrete random variables

04/05/2027 • B : 2 h – C : 1 h

Objectives of the session

· Calculate and interpret $E(X)$.

· Solve a binomial problem.

· Analyse your errors.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 100. 20

2. Give $E(X)$ for $X \sim \mathcal{B}(10, 0{,}3)$. 3

3. Calculate $\binom{5}{2}$. 10

4. Calculate the expectation of X constant equal to 5. 5

5. Express $X \geq 1$ by a complement. $1 - P(X = 0)$

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: calculate and interpret $e(x)$ and solve a problem binomial.

Investigation

A centre receives 120 groups of 10 customers, each customer complaining with probability 0 ,08. Estimate the mean number of complaints per group, then over all the groups. Discuss what the expectation does or does not allow us to predict.

Detailed solutions · teacher edition

By group $E(X) = 10 \times 0{,}08 = 0{,}8$. Over 120 groups, about 96 complaints on average; this is not a certainty. Have students say aloud the data that triggers the choice of method.

Written summary

For $X \sim \mathcal{B}(n, p)$, $E(X) = np$. The value gives the mean number of successes over a large number of series of n trials.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Detailed solutions · teacher edition

By group $E(X) = 10 \times 0{,}08 = 0{,}8$. Over 120 groups, about 96 complaints on average; this is not a certainty.

Practice · essentials

1. Justify that the number of successes among 7 independent trials, with $p = 0{,}3$, follows a binomial distribution.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduce $P(X \geq 1)$ and calculate $E(X)$.

Detailed solutions · teacher edition

$X \sim \mathcal{B}(7, 0{,}3)$. $P(X = 2) = \binom{7}{2} 0{,}3^2 0{,}7^5$, $P(X = 0) = 0{,}7^7$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{21}{10}$.

References in the textbook

Extensions: Calao, pp. 160-169. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the resources digital resources of the textbook.

Calao textbook · p. 166, Exercise 34

A company sells large quantities of low-energy, long-lifetime. 98 % of these bulbs have an operating life of at least 20,000 h. The bulbs are sold in batches of 10. We denote X the random variable which, to each batch of 10 bulbs, assigns the number of bulbs whose lifetime is at least 20,000 h. The number of bulbs is large enough for us to treat the the choice of a batch to a random sample, with replacement, of 10 light bulbs.

1. What is the distribution of X ? State its parameters.
2. Copy and complete: $\binom{10}{9} = \cdots$; $\binom{10}{10} = \cdots$.
3. Deduce the probabilities (round to 0,001) : $P(X = 9)$ and $P(X = 10)$. Interpret the results previous.
4. Deduce the probability (to 0,001) that a batch of 10 bulbs contains at least 9 bulbs whose lifetime will be at least 20,000 h.

Detailed solutions · teacher edition

$X \sim \mathcal{B}(10; 0{,}98)$, $\binom{10}{9} = 10$ and $\binom{10}{10} = 1$. $P(X = 9) = 10(0{,}98)^9(0{,}02) \approx 0{,}167$ and $P(X = 10) = 0{,}98^{10} \approx 0.817$. Thus $P(X \geqslant 9) \approx 0{,}984$.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 375 and the rate or common ratio by 5. Write four lines: data, formula, calculation, conclusion.

Detailed solutions · teacher edition

The solution depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method

Lesson plan for 1 h: 7 min of fluency; 10 min of sharing and written summary; 28 min of exercises essentials; 5 min of correction; 3 min of exit ticket.

Lesson plan for 2 h: add 20 min of activity work, 20 min of formalising the lesson, a second independent exercise and 10 min of comparative correction.

Differentiated teaching: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.